

MEFISTO is a novel computational framework designed to extend traditional factor analysis to datasets where samples are collected with known temporal or spatial relationships. In contrast to standard methods that assume sample independence, MEFISTO incorporates continuous covariates (such as time or spatial coordinates) directly into the model. Here are the key aspects of the method:

1. Integration of Multimodal Data and Continuous Covariates

- **Dimensionality Reduction:** Like conventional factor analysis, MEFISTO reduces high-dimensional data into a lower-dimensional latent space.
- **Multimodal and Multi-group Capability:** It is built to integrate data from multiple views (e.g., different omics) and groups (e.g., various species, conditions, or individuals).
- **Continuous Covariate Inclusion:** By incorporating time or spatial coordinates, it models dependencies that arise in longitudinal or spatial studies, thereby capturing patterns that are smooth over time or space.

2. Methodological Innovations

- **Gaussian Process Framework:** Each latent factor is modeled as a continuous process governed by a Gaussian process. This allows the method to flexibly capture both smooth (e.g., gradual developmental changes) and non-smooth variations (e.g., abrupt changes unrelated to time/space).
- **Data-driven Alignment:** When temporal (or spatial) sampling is imperfectly aligned between groups, MEFISTO employs dynamic time warping to align the covariate information, ensuring a coherent latent space across datasets.
- **Sparse Inference for Scalability:** The use of sparse Gaussian process approximations and efficient decompositions (like Kronecker products) enables the application of MEFISTO to large-scale datasets with many samples.

3. Practical Applications and Performance

- **Developmental Gene Expression Atlas:** The method has been applied to an evolutionary atlas of organ development, capturing conserved and species-specific developmental programs across multiple organs and species.
- **Longitudinal Microbiome Studies:** In sparse, time-resolved microbiome data from infants, MEFISTO distinguished trajectories linked to factors such as birth mode and diet, demonstrating robust imputation even with high missingness.
- **Single-cell Multi-omics and Spatial Transcriptomics:** MEFISTO has also been used to integrate different molecular layers (e.g., RNA, DNA methylation, chromatin accessibility) along multi-dimensional developmental trajectories, as well as to uncover spatial patterns in tissue sections.

4. Downstream Analysis and Interpretability

- **Imputation and Interpolation:** Once the latent factors are learned, the model can interpolate missing time or spatial points and provide uncertainty estimates.

- **Biological Insights:** The sparse loadings associated with each factor facilitate the interpretation of the underlying molecular or cellular processes, which can be further investigated via enrichment analyses.

In summary, MEFISTO addresses key challenges in analyzing complex, structured biological data by merging the strengths of factor analysis with a flexible Gaussian process framework. This approach not only improves factor recovery and imputation accuracy compared to non-temporal methods (such as MOFA) but also opens the door for more nuanced interpretation of spatio-temporal biological processes.

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The excerpt describes several key findings from the application of MEFISTO to diverse datasets:

Opossum-specific Factor 5:

- This factor highlights differences in gene expression trajectories that are unique to opossums.
- Specifically, its behavior (as shown in Supplementary Fig. 8c) is consistent with earlier observations of changes in the developmental timing of neutrophils in marsupials, suggesting that cell type composition in opossum tissues may differ from that in other species.

Handling Missing Data and Interpolation:

- The authors tested MEFISTO's robustness by intentionally masking random species–time point combinations.
- MEFISTO was able to interpolate completely missing time points (Supplementary Fig. 9) by leveraging both temporal covariate information and the correlations between features (Supplementary Fig. 10).

Application to Longitudinal Microbiome Data:

- In a study of infants' gut microbiomes, where the dataset is extremely sparse (with 91.4–98.0% zeros and many missing time points), MEFISTO identified distinct temporal trajectories:
 - **Factor 1** primarily captured variations linked to the mode of delivery (e.g., cesarean vs. vaginal), reflecting delayed microbiome maturation in cesarean-born infants.
 - **Factor 2** was associated with dietary differences, capturing sharp early changes in microbial composition.
- The model's factor values were robust even when data were masked, and the derived taxonomic enrichments (such as for *Faecalibacterium*, *Bacteroides*, *Clostridium*, and others) aligned well with known biological patterns of microbiome development.

Applications to Multi-dimensional and Spatial Omics:

- **Single-cell Multi-omics:**

- MEFISTO was applied to a dataset of 1,518 cells from early mouse development, profiled using combined nucleosome, methylation, and transcriptome sequencing (scNMT-seq).
 - The method identified factors that capture smooth developmental trajectories (e.g., corresponding to mesoderm and endoderm formation) as well as factors reflecting non-smooth biological processes, such as cell cycle differences.
 - Importantly, MEFISTO enabled the imputation of missing epigenetic measurements, enhancing the integration of these sparse data types.
- **Spatial Transcriptomics:**
 - When applied to a 10x Visium spatial transcriptomics dataset of the mouse brain, MEFISTO successfully uncovered spatial patterns corresponding to major anatomical regions.
 - The analysis highlighted specific marker genes (for example, *Ttr* for the choroid plexus) and provided an integrated measure of spatial smoothness, demonstrating its utility for spatially resolved data.

Overall, these results illustrate that MEFISTO not only improves factor recovery and imputation in datasets with high missingness but also effectively disentangles both shared and group-specific spatio-temporal patterns across a range of biological contexts. This flexibility makes it a powerful tool for analyzing complex, multimodal, and structured biological data.

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The Methods section describes in detail how MEFISTO is formulated and implemented. Here's a structured summary:

1. Model Formulation

- **Data Structure:**
 - MEFISTO takes as input a collection of matrices $Y_{m,(g)}$ where each matrix corresponds to a specific group (g) and view (m).
 - Each matrix contains $N_{(g)}$ samples (rows) and D_m features (columns).
 - Every sample is additionally associated with a continuous covariate ($C_{(g)}$) representing, for example, time or spatial coordinates.
- **Matrix Decomposition:**
 - The data are jointly decomposed as:

$$Y_{m,(g)} = Z_{(g)}W_m^T + \varepsilon_{m,(g)}$$
 - Here, $Z_{(g)}$ (size $N_{(g)} \times K$) contains the latent factors and W_m (size $D_m \times K$) contains the feature-specific weights.
 - A sparsity prior is imposed on W_m to aid interpretability.
- **Incorporating the Covariate with Gaussian Processes:**

- Each latent factor value (z_{nk}) is modeled as a realization of a Gaussian process (GP):

$$z_{nk} = f_k(c_n) + \eta_n$$
with $f_k \sim \text{GP}(0, \kappa_k)$.
- The covariance function κ_k is designed to capture both intergroup relationships and the smoothness along the covariate:

$$\kappa_k(c_n, c_l) = \kappa_k^G(g, h) \cdot \kappa_k^C(c_n, c_l).$$
- A low-rank covariance function is used for group relationships (κ_k^G), and a squared exponential function is used for the covariate (κ_k^C).
- **Hyperparameters:**
 - The hyperparameters (such as x_k , σ_k , l_k , and s_k) determine:
 - The covariance structure between groups.
 - The smoothness and the scale of variation along the covariate.
 - The parameter s_k balances smooth versus non-smooth variation, while l_k controls how quickly correlations decay along the covariate.

2. Inference and Computational Strategies

- **Variational Inference:**
 - MEFISTO uses a variational approach to optimize the evidence lower bound (ELBO) and infer both the latent factors and the GP hyperparameters.
- **Scalability:**
 - For large datasets, the method can use a subset of covariate values (via a regular grid) to reduce computational load.
 - When the covariance matrix decomposes into a Kronecker product ($\text{KG} \otimes \text{KC}$), MEFISTO exploits this structure for faster inference using spectral decompositions.

3. Alignment of Misaligned Data

- When samples across groups are not perfectly aligned in time or space, MEFISTO learns a non-linear warping function (using dynamic time warping) in the latent space.
- This alignment is performed by interleaving updates of the model parameters with an optimization step that minimizes differences between groups relative to a reference.

4. Data Preprocessing and Model Setup

- Different likelihood models can be applied per view, similar to previous multimodal factor analysis frameworks.
- Preprocessing (such as handling over-dispersion or normalizing count data) is recommended to tailor the input to the characteristics of the data.

5. Downstream Analyses

- Once trained, the model allows:
 - **Interpolation/Extrapolation:** Predicting latent factors for new covariate values, which is useful for imputing missing data.
 - **Imputation:** Missing feature values are inferred via the model equation.
 - **Interpretability:** The inferred hyperparameters provide insights into the smoothness of each factor and the intergroup relationships, which can be used for clustering or outlier detection.

6. Simulation Studies and Applications

- **Simulations:**
 - Data simulated from the generative model were used to assess factor recovery and imputation performance, comparing MEFISTO with methods like MOFA and univariate Gaussian processes.
- **Real Data Applications:**
 - The model was applied to diverse datasets such as evolutionary gene expression (evo-devo), sparse longitudinal microbiome studies, single-cell multi-omics, and spatial transcriptomics.
 - These applications demonstrate MEFISTO's ability to capture both shared and group-specific patterns, handle missing data, and align misaligned temporal or spatial profiles.

7. Related Methods and Context

- The section contrasts MEFISTO with other factor analysis and tensor decomposition methods, noting that many existing approaches do not incorporate continuous side-information or handle misaligned time courses and intergroup heterogeneity as effectively.

This comprehensive methodology enables MEFISTO to extract meaningful latent factors from complex, multimodal, and temporally or spatially structured data while addressing challenges like missing values and misalignment between groups.

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This notebook provides a comprehensive demonstration of how to apply MEFISTO to EvoDevo data to identify temporal variation across species and organs. The workflow includes:

1. **Data Preparation:**
 - Loading a normalized gene expression dataset covering 5 species (groups) and 5 organs (views) along with developmental time information.
 - Converting the data into AnnData objects for each view and then assembling these into a multimodal MuData object.
2. **Model Execution with MEFISTO:**

- Running MEFISTO via the `mu.tl.mofa` function, where the developmental time is used as a continuous covariate (via the 'time' column) and species serve as groups.
- Enabling smooth warping of time points (with “Mouse” as the reference) allows the model to learn an alignment of developmental stages across species.
- The model leverages the underlying Gaussian process framework to interpolate latent factors for new, unseen time points.

3. Visualization and Analysis:

- Visualization of the learned latent factor space with scatter plots colored by species, raw time, and warped time.
- Comparison of factor trajectories before and after alignment, highlighting the benefit of aligning developmental time across species.
- Downstream analyses using the `mofax` library to visualize factor interpolation, assess the smoothness of each latent factor, and examine group (species) kernel matrices to evaluate sharedness across species.

For more detailed information about MEFISTO, you can visit its [website](#) or consult the [preprint](#) by Britta Velten et al.

This notebook serves as a practical guide to integrating multimodal omics data with temporal covariates using MEFISTO, showcasing its ability to handle complex datasets with missing values and misaligned time points, and to extract biologically meaningful latent factors.

This notebook illustrates how to leverage MEFISTO for identifying temporal variation in evolutionary developmental (EvoDevo) data by integrating multimodal gene expression measurements across different organs and species. The key steps are:

1. Data Loading and Preparation:

- The notebook starts by reading in a normalized gene expression dataset for 5 species (e.g., Human, Mouse, Opossum, Rabbit, Rat) and 5 organs (Brain, Cerebellum, Heart, Liver, Testis), along with associated developmental time points.
- It then creates separate `AnnData` objects for each organ (view) and combines them into a single multimodal `MuData` object.
- Sample metadata (including the developmental time and species labels) is joined to the `MuData` object to provide the continuous covariate and grouping information.

2. Running MEFISTO:

- The MEFISTO analysis is invoked via the `mu.tl.mofa` function. Here, the 'time' column is specified as the covariate, and 'species' is used as the grouping variable.

- Smooth warping is enabled (using "Mouse" as the reference) so that the model can learn an alignment of developmental time across different species, even when time points are misaligned.
- The Gaussian process framework underlying MEFISTO allows the model not only to learn smooth latent factors but also to interpolate factor values at new (unseen) time points.

3. Visualization of Results:

- The learned latent factors are visualized in several ways:
 - Scatter plots of the factor space are colored by species, raw time, and the warped (aligned) time, demonstrating how alignment brings different species onto a common developmental trajectory.
 - The notebook compares the relationship between a specific factor (e.g., Factor1) and the raw versus warped time, highlighting the effect of alignment.
 - Additionally, the alignment itself is visualized by plotting raw time against warped time, showing how the developmental time scales have been adjusted across species.

4. Further Analysis with mofax:

- Using the mofax library, the notebook further examines the model by:
 - Interpolating latent factors to predict values at intermediate or missing time points.
 - Visualizing the smoothness of each latent factor, which reflects how smoothly a developmental process unfolds.
 - Assessing the sharedness of the factors across species via the learned group kernel, providing insight into how similarly different species follow a given developmental trajectory.

Overall, this workflow demonstrates how MEFISTO uses continuous covariate information (developmental time) to perform multimodal factor analysis, align misaligned time courses, and extract interpretable latent factors that capture biologically relevant temporal variation across species. For additional details on the method, you can refer to the [MEFISTO website](#) and the [preprint by Britta Velten et al.](#)

This notebook demonstrates how to apply MEFISTO to longitudinal microbiome data to capture temporal dynamics in infants' gut microbiomes. The workflow is as follows:

1. Data Loading and Preparation:

- The notebook loads a preprocessed microbiome dataset (microbiome_data.csv) along with sample metadata (including group, month, delivery mode, diet, and sex) and feature metadata (taxonomic information).

- An AnnData object is created by pivoting the data into a sample-by-feature matrix, and sample metadata is joined to the AnnData object. Feature-level taxonomic annotations are also added to aid downstream interpretation.

2. Running MEFISTO:

- MEFISTO is executed using the `mu.tl.mofa` function on the AnnData object, where the metadata column 'month' is used as the continuous covariate to model temporal variation.
- The grouping variable 'group' (representing different children) is specified so that the model can learn both shared and group-specific temporal patterns.
- The smooth covariate framework is activated, and options are provided to optimize the latent space (e.g., grid size and optimization frequency).

3. Visualization of the Latent Factors:

- The learned latent factors are visualized using scatterplots. For example, Factor1 is plotted against the month of life and colored by delivery mode or diet, showing how the microbiome evolves over time in different conditions.
- These visualizations help illustrate the temporal trends captured by the latent factors.

4. Downstream Analysis with mofax:

- The pre-trained MEFISTO model is loaded using the Python package `mofax`, which offers convenient plotting functions.
- Variance decomposition is visualized to see how much variance each factor explains in the microbiome data.
- The factors are plotted against the month of life, with samples colored by metadata (such as delivery mode or diet), to reveal distinct temporal trajectories.
- Factor weights are examined to identify the microbial features most strongly associated with a particular factor (e.g., Factor1).
- The notebook further aggregates these weights at the genus level, highlighting, for instance, that species of *Bacteroides* and *Faecalibacterium* (with negative Factor1 weights) increase in abundance in vaginally delivered children.

Overall, the notebook provides a practical example of using MEFISTO to integrate longitudinal microbiome data, allowing researchers to (i) model temporal changes in a high-dimensional dataset, (ii) compare developmental trajectories across groups (children), and (iii) link latent factors to biologically relevant features such as microbial taxa. More details about the method can be found on the [MEFISTO website](#) and in the [preprint by Britta Velten et al.](#)

This notebook demonstrates how to apply MEFISTO to spatial transcriptomics data by leveraging the spatial coordinates as continuous covariates. Key steps include:

1. Data Loading and Preprocessing:

- The spatial transcriptomics dataset ("Mouse Brain Serial Section 1 (Sagittal-Anterior)" from 10X Genomics) is loaded using Scanpy's `read_visium` function.
- Standard quality control and filtering steps are performed (e.g., calculating QC metrics, filtering based on total counts and mitochondrial percentage).
- The spatial coordinates (`imagerow` and `imagecol`) are extracted from the data and added to the observation metadata for easy access during modeling.

2. Training the MEFISTO Model:

- The model is run via `mu.tl.mofa`, specifying a smooth covariate defined by the two spatial dimensions (`[imagerow, imagecol]`).
- A sparse Gaussian process is used—with 1000 inducing points—to efficiently learn spatial covariance patterns in the data.
- The model is trained for several iterations while monitoring the evidence lower bound (ELBO), which shows how the model converges over time.

3. Downstream Analyses and Visualization:

- After training, the MEFISTO model is loaded using the `mofax` package.
- Factor correlation plots are generated to verify that the latent factors are uncorrelated.
- Spatial patterns of the latent factors are visualized by overlaying the factor values on the tissue image using Scanpy's spatial plotting functions.
- The smoothness of each factor is examined via bar plots, reflecting how smoothly each latent process varies across space.
- For example, Factor4 is further analyzed by extracting its top weighted genes (e.g., `Ttr`, `Enpp2`, `Igfbp2`) and visualizing their spatial expression patterns to reveal biologically relevant spatial gradients.

Overall, the notebook illustrates how MEFISTO integrates high-dimensional spatial transcriptomics data with spatial coordinates to capture smooth latent factors that reflect spatial variation in gene expression. For more information about MEFISTO, please refer to its [website](#) and the [preprint by Britta Velten et al.](#)

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Main Concepts of MEFISTO

MEFISTO (Multimodal Factor Analysis with Spatio-Temporal Optimization) is a computational framework designed to identify latent patterns in multimodal datasets with temporal or spatial dependencies. Its core concepts include:

1. Multimodal Integration:

- Combines data from multiple "views" (e.g., gene expression, microbiome, methylation) and "groups" (e.g., species, individuals) into a unified latent space.
- Each latent factor captures shared or group-specific variation across modalities.

2. Spatio-Temporal Modeling:

- Uses **Gaussian Processes (GPs)** to model latent factors as smooth functions of continuous covariates (time or space).
- Captures both smooth (gradual) and non-smooth (abrupt) variation, distinguishing biological processes from noise.

3. Alignment of Misaligned Data:

- Applies **dynamic time warping (DTW)** to align temporal trajectories across groups (e.g., developmental stages of different species).
- Handles irregular sampling and missing data via GP-based interpolation.

4. Scalability and Robustness:

- Leverages **sparse GP approximations** and **Kronecker decompositions** for efficient computation on large datasets.
- Provides uncertainty estimates for predictions and imputations.

5. Interpretability:

- **Sparsity priors** on feature weights highlight key drivers of each factor.
- Metrics like **smoothness** (how smoothly a factor varies) and **sharedness** (cross-group similarity) quantify factor properties.

Method Explanation

1. Model Structure:

- **Input:** Matrices for each view (e.g., organ-specific gene expression) and group (e.g., species), with samples annotated by covariates (time/space).
- **Decomposition:**

$$Y_{m,g} = Z_g W_m T + \epsilon_{m,g}, Y_{m,g} = Z_g W_m T + \epsilon_{m,g},$$

where $Z_g Z_g^T$ (latent factors) and $W_m W_m^T$ (feature weights) are learned.

- **GP Integration:** Each factor $z_{n,k}$ is modeled as $z_{n,k} = f_k(c_n) + \eta_n z_{n,k} = f_k(c_n) + \eta_n$, where $f_k \sim \text{GP}(0, k)$. The covariance kernel k captures group relationships and covariate smoothness.

2. Inference:

- **Variational Inference:** Optimizes the evidence lower bound (ELBO) to infer factors, weights, and GP hyperparameters.
- **Alignment:** Uses DTW to align misaligned covariates (e.g., developmental time across species).

3. Downstream Analysis:

- **Interpolation:** Predicts latent factors at unobserved time/space points.
- **Visualization:** Overlays factors on spatial maps or temporal trajectories.
- **Enrichment:** Links factors to biological features (e.g., genes, microbes) via sparsity-driven weights.

Advancements Over State-of-the-Art Methods

1. Explicit Covariate Modeling:

- Unlike traditional factor models (e.g., PCA, MOFA), MEFISTO directly incorporates time/space via GPs, capturing smooth trends without post-hoc analysis.
- **Example:** MOFA treats samples as independent, leading to inaccurate factor recovery in time-series data.

2. Alignment of Heterogeneous Data:

- DTW aligns misaligned covariates (e.g., developmental stages across species), a feature absent in methods like MOFA or tensor decomposition.

3. Robustness to Sparsity:

- GP interpolation handles missing data better than non-temporal methods, which treat samples as independent.
- **Validation:** MEFISTO outperformed MOFA in imputing missing values in simulated and real datasets.

4. Interpretable Outputs:

- Smoothness and sharedness scores quantify factor dynamics, while sparse weights highlight biologically relevant features.

Applications on Diverse Datasets

1. Evolutionary Developmental Biology (EvoDevo)

- **Data:** Gene expression across 5 organs (views) and 5 species (groups) at developmental time points.
- **Key Findings:**
 - **Conserved Factors:** Gradual changes in proliferation genes (e.g., *IGF2BP1*) across species.
 - **Species-Specific Factors:** Human-specific *TRPM8* in liver, opossum-specific neutrophil markers.
- **Alignment:** DTW aligned developmental timelines, enabling cross-species comparison.
- **Impact:** Revealed evolutionary conservation and divergence in organ development.

2. Longitudinal Microbiome Study

- **Data:** Infant gut microbiome profiles (969 microbes) over 24 months, with metadata on delivery mode and diet.
- **Key Findings:**
 - **Factor 1:** Cesarean-born infants showed delayed colonization of *Bacteroides* (vaginal delivery marker).

- **Factor 2:** Formula feeding correlated with *Clostridiales* enrichment.
- **Robustness:** Handled 91-98% sparsity and irregular sampling.
- **Impact:** Linked clinical variables to microbiome maturation trajectories.

3. Spatial Transcriptomics

- **Data:** Mouse brain gene expression (10X Visium) with 2D spatial coordinates.
- **Key Findings:**
 - **Factor 4:** Captured choroid plexus region via *Ttr* expression.
 - **Smoothness:** High smoothness scores reflected continuous spatial patterns (e.g., cortical layers).
- **Efficiency:** Sparse GP with 1,000 inducing points reduced runtime without sacrificing accuracy.
- **Impact:** Mapped gene expression gradients to anatomical structures without single-cell references.

Conclusion

MEFISTO bridges the gap between traditional factor analysis and spatio-temporal modeling, offering a unified framework for multimodal data integration. Its ability to align heterogeneous timelines, interpolate missing data, and extract interpretable latent factors makes it indispensable for developmental biology, microbiome research, and spatial omics. By explicitly modeling time/space, MEFISTO outperforms methods like MOFA in accuracy and biological relevance, empowering researchers to uncover hidden patterns in complex datasets.